

SMART INDIA HACKATHON 2026



TITLE PAGE

PROBLEM STATEMENT ID

SIH26171

PROBLEM STATEMENT TITLE

On-device Visual Perception for Light-weight
Browser Agents

THEME & CATEGORY

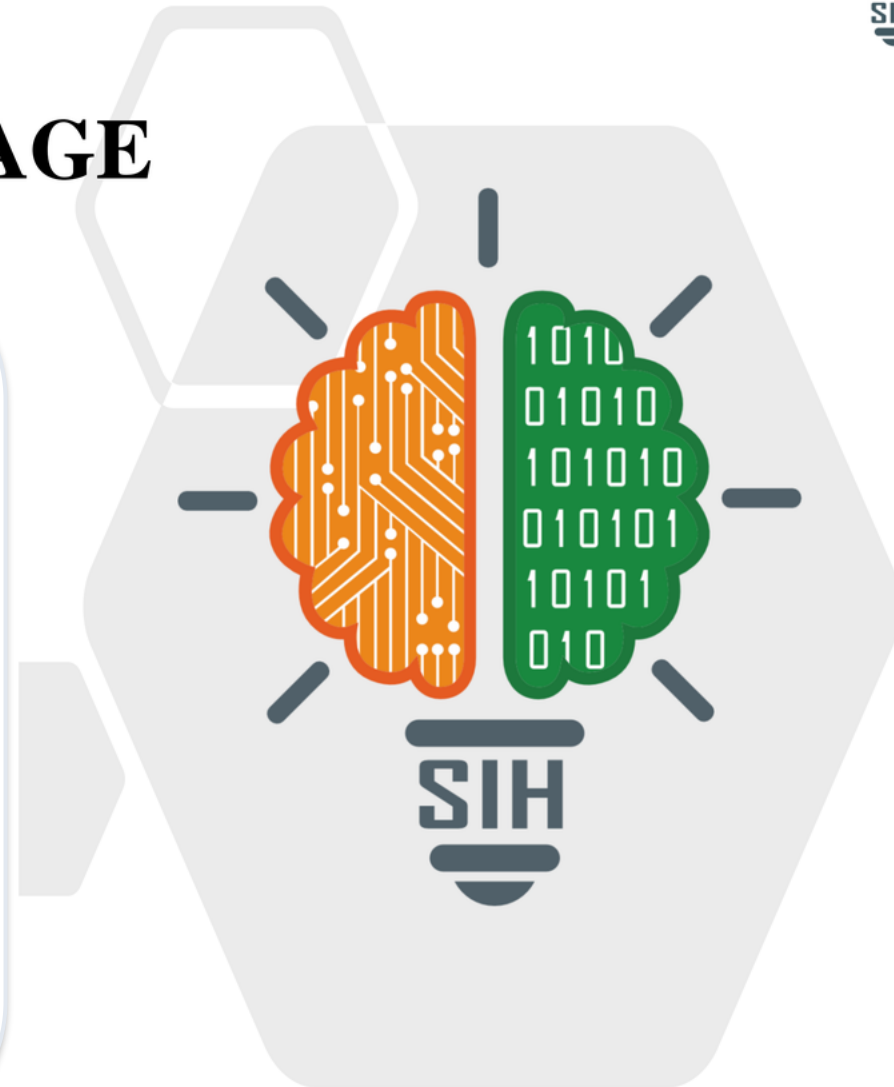
Smart Automation | Software

SUBMITTING ORGANIZATION

ISRO / Department of Space

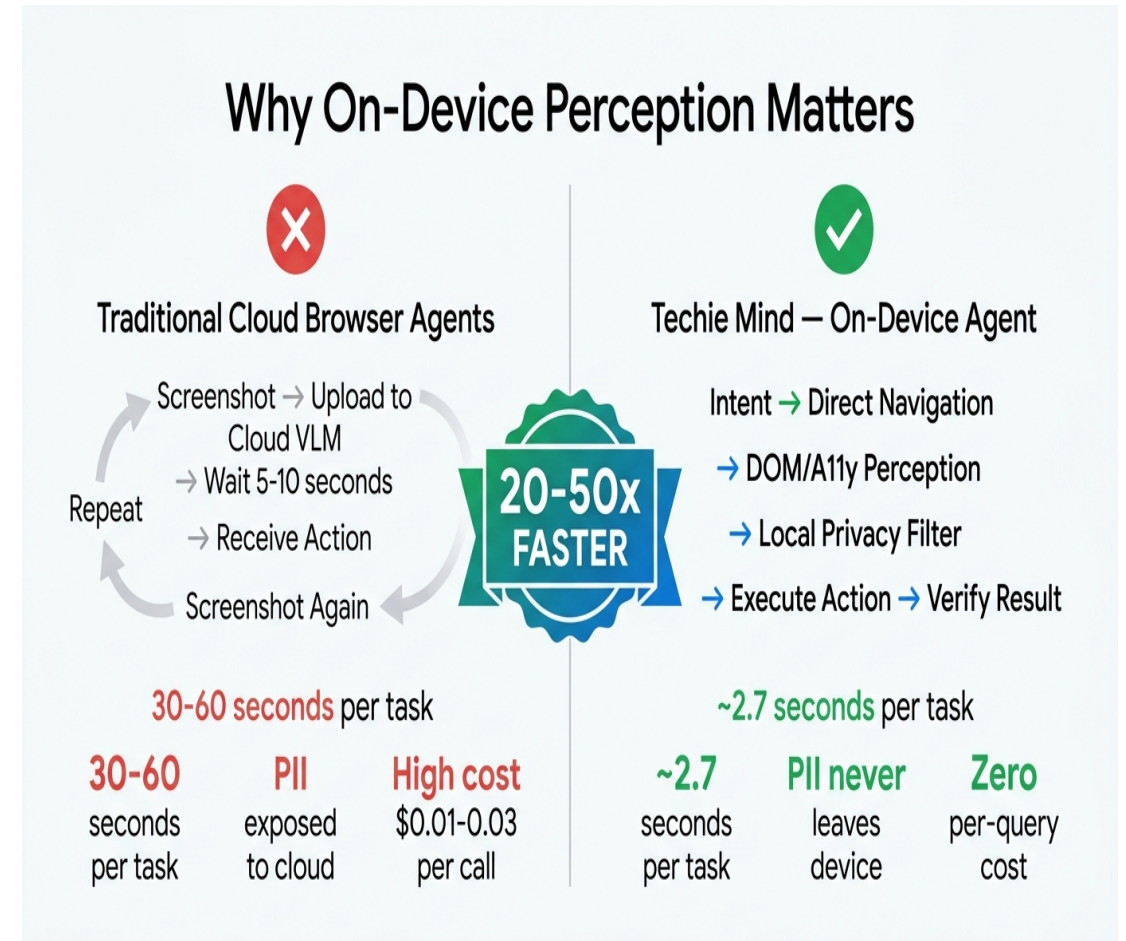
TEAM DETAILS

Team ID: 169984 | Team Name: Techy Guys



PROPOSED SOLUTION: On-Device Autonomous Browser Agent with Zero-Cloud Privacy

- **Techie Mind Overview:** Autonomous browser agent running directly inside Chrome/Firefox without cloud dependencies or per-query API bills.
- **4-Level Local Perception Stack:** Hierarchical cascade: DOM Observer → Accessibility Tree → Semantic Grounding → Local Florence-2 VLM (runs via WebGPU only when needed).
- **Local Privacy & Redaction Engine:** Detects & redacts Aadhaar, PAN, phone, passwords, and credit cards in memory. Raw sensitive data NEVER leaves user's computer.
- **Action Firewall Architecture:** Separates AI reasoning from execution authority. Model only proposes; local engine verifies schema, page state, and safety constraints.
- **Human Handover Protocol:** Payments, OTPs, and CAPTCHAs strictly paused for user control — preventing unauthorized transactions.
- **Vernacular Multi-Lingual Support:** Understands commands in English, Hindi, Kannada, Tamil, Telugu, and Hinglish via text and voice.
- **20–50x Faster than Cloud VLMs:** Typical task execution completes in ~2.7s (vs 30–60s on cloud VLM screenshot loops).

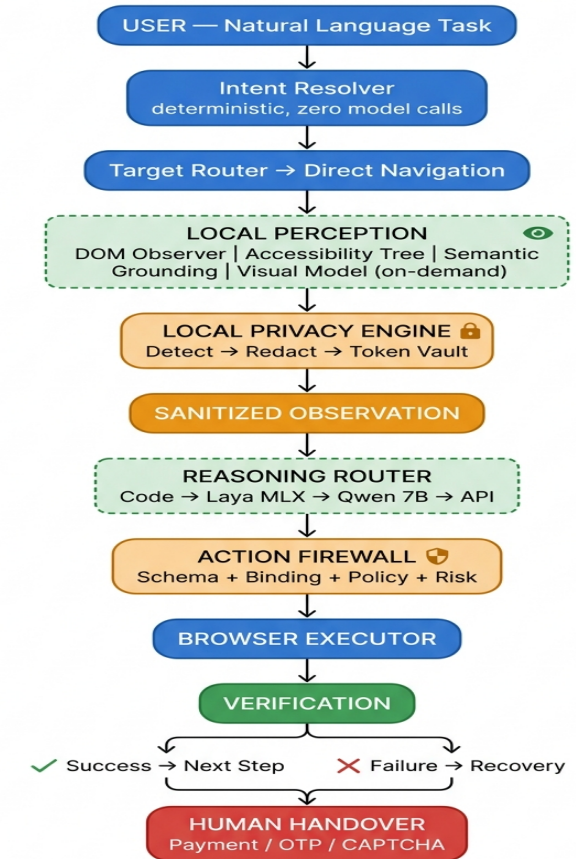


METHODOLOGY & ARCHITECTURE: 6-Step Deterministic-First Agent Pipeline

- **Step 1 — Intent Resolution:** Deterministic parsing extracts action and parameters in 5.3 ms with 0 AI model calls.
- **Step 2 — Direct Navigation:** Navigates directly to target domains without searching via Google, saving 2-3 round-trips.
- **Step 3 — Local Perception Cascade:** Extracts interactive DOM + A11y tree first; invokes Florence-2 ONNX VLM via WebGPU only if visual ambiguity exists.
- **Step 4 — Hardware-Boundary Privacy:** PII detector redacts credentials & sensitive tokens into a secure, TTL-limited in-memory vault.
- **Step 5 — Action Firewall Validation:** Validates proposed action against Zod runtime schema, target binding, and risk classification rules.
- **Step 6 — Execution & Verification:** Dispatches synthetic events, inspects post-action DOM state, and engages a 6-level recovery ladder upon failure.

Runtime	Perception	Reasoning	Validation
Chrome MV3 / Firefox	DOM + A11y Tree	Deterministic Code	Zod Schemas (Runtime)
TypeScript Strict	Florence-2 (WebGPU)	Laya MLX (Local)	Playwright Chromium
Preact (~3KB UI)	WAI-ARIA Grounding	Qwen 2.5:7B (Ollama)	Vitest Unit Test Suite

Techie Mind — System Architecture



FEASIBILITY & EMPIRICAL BENCHMARKS: Verified Live on Real Chromium (YouTube & Flipkart)

Implementation & Testing Rigor:

- ✓ Phase 0 (Foundation) & Phase 1 (Agent Core) 100% COMPLETE & PASSING.
- ✓ 189 Unit Tests + 21 Real-Chromium Browser Tests + 9 Live-Site Tests.
- ✓ Tested live on YouTube (Kannada/English search) and Flipkart (Product search).

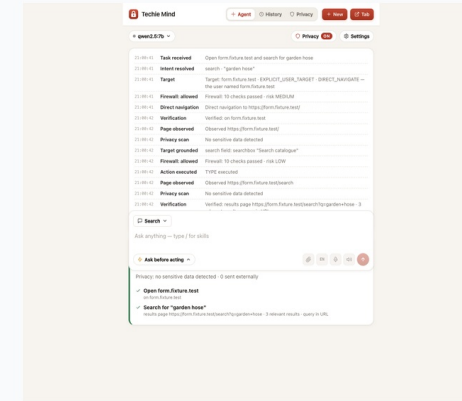
Measured Live Execution Benchmarks (Real Data):

Pipeline Stage	Median Latency
Intent Resolution (NL → Structured)	5.3 ms
Navigation & Page Readiness	1,533.0 ms
DOM & Accessibility Observation	38.0 ms
Semantic Element Grounding	0.4 ms
Action Dispatch & Execution	10.8 ms
Post-Action State Verification	1.9 ms
TOTAL END-TO-END (0 Model Calls)	~2,731 ms (2.7s)

Key Mitigations: WebGPU unsupported devices fall back seamlessly to WebAssembly; dynamic websites handled by semantic grounding without brittle selectors.

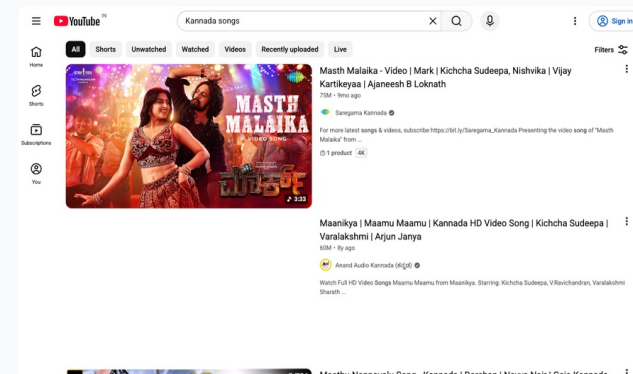
Live Agent Timeline (Task Finished in 1088 ms · 0 Model Calls)

Deterministic execution verified on live Chromium with zero external cloud dependencies



Real-World Automation on YouTube (Kannada Songs Voice/Text Search)

Shows automated DOM typing overlay with live visual verification badge



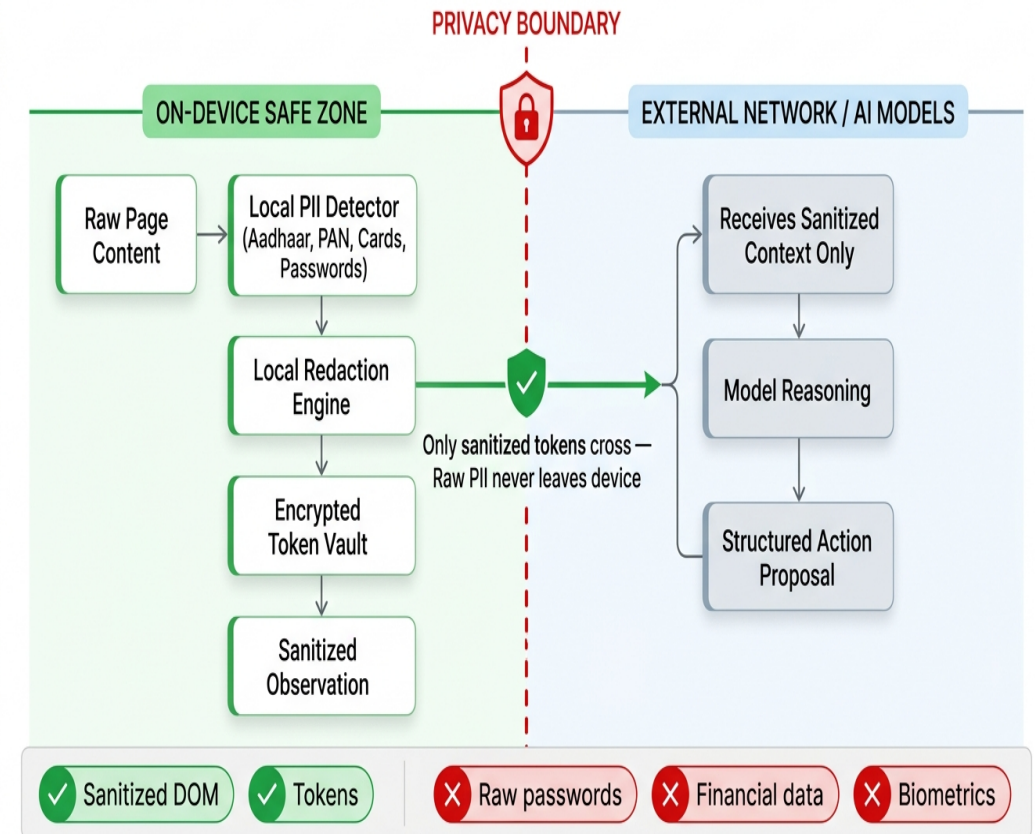
IMPACT & BENEFITS: Digital Inclusion, Data Sovereignty, and Multi-Dimensional ROI

Target Audience & Societal Impact:

- **Differently-Abled Users:** Hands-free web browsing via voice commands combined with Accessibility Tree perception.
- **Vernacular Indian Users:** Voice & text navigation in 6+ Indian languages removes digital literacy barriers.
- **National Data Sovereignty:** Zero-cloud processing guarantees sensitive data never leaves Indian borders.

Comprehensive Value Dimensions:

Dimension	Strategic Benefit
 Privacy	Hardware-enforced redaction prevents Aadhaar & financial leaks
 Economic	Zero per-query cost (cloud VLM agents cost \$0.01-\$0.03/step)
 Performance	20-50x faster execution (~2.7s vs 30-60s on cloud agents)
 Environmental	No power-hungry cloud GPU roundtrips, lowering carbon emissions
 Strategic	Demonstrates cutting-edge on-device AI developed for Indian users



TECHNOLOGY STACK: 12 Open-Source, Production-Grade Technologies Across Runtime, AI & Testing

TECHNOLOGIES & ARCHITECTURE STACK

Core Platform & Extension



TypeScript
Type Safety



Chrome MV3
Extension API



Firefox WebExtension
Cross-Browser



Preact
Fast UI

On-Device AI & Models



Florence-2 VLM
Vision-Language



Transformers.js
Browser ML



Qwen 2.5 7B
On-Device LLM



WebGPU
Hardware Acceleration

Infrastructure & Testing



Zod Contracts
Data Validation



Vite Bundler
Fast Builds



Vitest Runner
Unit Testing



Playwright Real-Browser
E2E Testing

100% On-Device Execution: All perception models and reasoning pipelines run locally without sending page context to cloud servers.

RESEARCH & REFERENCES: Grounded in State-of-the-Art Vision-Language & Web Standards

1. **SIH Problem Statement SIH26171:** On-device Visual Perception for Light-weight Browser Agents (ISRO / Dept. of Space)
2. **Microsoft Research (2023):** Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks (arXiv:2311.06242)
3. **Hugging Face Transformers.js:** State-of-the-art machine learning in the browser via ONNX Web
4. **W3C WebGPU Specification:** Next-generation browser acceleration standard for on-device ML compute
5. **Alibaba Cloud Qwen 2.5:** Open-weight foundational language models for lightweight reasoning
6. **Apple Machine Learning:** MLX framework for efficient local model inference
7. **Google Chrome Extensions MV3:** Manifest V3 architecture, service workers, and native sidepanel APIs
8. **W3C Accessible Name & Description (AccName 1.2):** Accessible tree computation for semantic browser interaction
9. **Zod Runtime Schema Validation:** Type-safe runtime contracts and schema boundaries
10. **Microsoft Playwright:** Cross-browser end-to-end automated testing across real Chromium & WebKit

Techie Mind and Resarch-& References

